Free Shipping Threshold Experiment — Result Review

This section reviews the synthetic experiment results using the same framework typically used by experimentation teams at companies like **Google, Airbnb, Netflix, and Meta**.

The review proceeds in the following order:

1. Diagnostics (experiment health)
2. Primary metric evaluation
3. Mechanism metrics
4. Guardrail metrics
5. Decision rule application

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1. Diagnostics (Experiment Health)

The first step in any experiment review is verifying that the experiment ran correctly.

The key question is:

> Can we trust the data?

Sample Ratio Mismatch (SRM)

Expected traffic allocation:

Arm	Expected Sessions
t35	~150,000
t50	~150,000
t65	~150,000

Observed sessions:

Arm	Sessions
t35	149,618
t50	150,306
t65	149,618

SRM test result:

```
p-value = 0.3489
```

Interpretation:

- p-value > 0.05
- traffic allocation appears random
- **no SRM detected**

This indicates the experiment assignment system behaved correctly.

Visitor Assignment Consistency

```
inconsistent_test_assignments: 0
```

Meaning:

- each visitor saw only one threshold
- visitor-level randomization worked correctly

Data Logic Validation

All validation checks passed:

```
shipping_logic_valid: True
total_revenue_formula_valid: True
contribution_margin_formula_valid: True
```

This confirms:

- shipping rules were applied correctly
- revenue formulas are correct
- contribution margin calculations are correct

These checks must pass before interpreting experiment results.

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2. Primary Metric Review

The primary decision metric is:

```
Contribution Margin per Session
```

Results:

Arm	CM per Session

Arm	CM per Session
t35	0.232
t50	0.332
t65	0.334

Interpretation:

- **t35 performs substantially worse**
- **t50 and t65 perform similarly**
- **t65 is slightly higher**

Difference between the top two arms:

```
t65 − t50 ≈ +0.002
```

This difference is extremely small.

In a real experiment review, the immediate question would be:

> Is this difference statistically significant or economically meaningful?

Most likely, this difference is **not economically meaningful**.

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3. Mechanism Metrics

Mechanism metrics help explain **why the experiment produced these results**.

Conversion Rate

Arm	Conversion Rate
t35	2.21%
t50	2.05%
t65	1.75%

Interpretation:

Lower shipping thresholds reduce checkout friction and increase conversion.

Observed pattern matches expectations.

Average Order Value (AOV)

Arm	AOV
t35	44.4
t50	55.4
t65	60.0

Interpretation:

Higher thresholds encourage customers to add items to qualify for free shipping.

Observed pattern matches expectations.

Revenue per Session

Arm	Revenue per Session
t35	1.02
t50	1.20
t65	1.12

Interpretation:

- t35 loses revenue due to smaller baskets
- t65 loses revenue due to lower conversion
- t50 balances both effects

This is a classic free-shipping threshold tradeoff.

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4. Guardrail Metrics

Guardrail metrics ensure that improvements in the primary metric do not harm the business.

Free Shipping Qualification Rate

Arm	Qualification Rate
t35	71%
t50	49%
t65	33%

Interpretation:

Lower thresholds lead to more free shipping.

This increases subsidy exposure.

Shipping Subsidy per Order

Arm	Subsidy per Order
t35	7.28
t50	5.95
t65	4.96

Interpretation:

Lower thresholds significantly increase shipping subsidy.

Higher thresholds reduce subsidy exposure.

Negative Margin Orders

```
0% across all arms
```

This is acceptable for synthetic data.

In real ecommerce systems, this metric typically ranges between **1–5%**.

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5. Decision Rule Application

The experiment decision rules state that a new threshold should be adopted only if:

1. Contribution margin per session improves
2. Revenue per session is not materially harmed
3. Guardrail metrics remain within acceptable ranges

Evaluation of t35

Contribution margin per session:

```
0.232 vs baseline 0.332
```

Result:

Significant decline.

Decision:

❌ Reject t35

Evaluation of t65

Contribution margin per session:

```
0.334 vs baseline 0.332
```

Small improvement.

Revenue per session:

```
1.12 vs baseline 1.20
```

Revenue declines.

Guardrails:

- shipping subsidy improves
- conversion declines

Final Decision

Recommended policy:

```
Maintain the current $50 free shipping threshold.
```

Reasoning:

- t35 materially reduces contribution margin
- t65 provides only a negligible improvement in margin
- t65 reduces revenue per session and conversion
- t50 provides the best balance between conversion and basket size

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Why This Is a Good Synthetic Experiment

This experiment produces a realistic outcome.

Many real-world experiments confirm that the **existing policy is already near optimal**.

Observed pattern:

Arm	Outcome
t35	too generous
t50	balanced
t65	too restrictive

This result demonstrates a common experiment outcome:

```
Current policy ≈ optimal policy
```

Recommended Additional Analysis

Before producing the final experiment readout, two additional analyses should be performed.

Confidence Intervals

Compute confidence intervals for:

- contribution margin per session
- revenue per session
- conversion rate

This helps quantify uncertainty in the estimates.

Treatment Effect Table

Example summary:

Comparison	Δ CM per Session
t35 vs t50	-0.10
t65 vs t50	+0.002

This table makes the effect sizes easier to interpret.

Final Assessment

The synthetic experiment generator produced:

- realistic behavioral tradeoffs
- believable economic relationships

- a non-obvious outcome

This makes the experiment suitable for demonstrating a full experimentation workflow.